Imran Ali Shah

PhD in Biomedical Engineering (Pending Conferral)

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https://eliteheat.github.io/eliteheatt.github.io/

Education

Georgia Institute of Technology & Emory University

PhD in Biomedical Engineering — 2020 – 2026

- Advisors: Lakhsmi P. Dasi, Alessandro Veneziani
- Cumulative GPA: 4.0
- All Requirements Completed
- Degree Conferral: May 2026

Georgia Institute of Technology

B.S. in Biomedical Engineering — 2016 – 2018

- Cumulative GPA: 3.80, Major GPA: 3.63

Research Interests

- Cardiovascular Biomechanics, Congenital Heart Defects, Valvular Heart Diseases, Coronary Artery Disease, Computational Fluid Dynamics, Fluid-Structure Interaction, Scientific Computing, Reduced Order Modeling,

Research Experience

American Heart Association Predoctoral Fellow — 01/2024 – Current

Emory University

- Development of a fast predictive framework of transcatheter aortic valve replacement (TAVR) deployment procedures via reduced order modeling and surrogate contact methods, enabling patient-specific mechanical insights into the risk of coronary obstruction following TAVR in a real-time setting (https://doi.org/10.58275/AHA.24PRE1194863.pc.gr.190665)

Graduate Research Assistant — 08/2020 – Current

Cardiovascular Fluid Mechanics Laboratory, Emory Center for Mathematics and Computing in Medicine

Advisors: Lakshmi P. Dasi, Alessandro Veneziani

- Developing model reduction techniques for real-time predictive computational models of Transcatheter Aortic Valve Replacement (TAVR) procedures
- Developing model reduction framework for real-time shape optimization of the total cavopulmonary connection for Fontan surgical planning
- Investigating the effects of drug-eluting stent platforms (Medtronic Resolute and Abbott Xience DES) on the endothelial wall shear stress in patient-specific coronary arteries
- Application of model reduction techniques and computational mechanics for fluid-structure interaction (FSI) analysis in post-TAVR deployment

Patient-Specific Fontan Surgical Planning — 06/2021 – Current

- Coordinating with clinicians and surgeons from Children's Hospital of Atlanta, Children's Hospital of Philadelphia, and Nationwide Children's Hospital for optimizing Fontan surgeries for children with congenital heart defects
- Creating geometrical models and virtual surgical planning procedures for patient-specific Fontan cases from magnetic resonance imaging (MRI) and computed tomography (CT) scans
- Performing computational fluid dynamics (CFD) simulations for several surgical options for each individual patient and identifying the most optimal surgical option in each scenario

Publications and Original Works

- Travis Singh, Stephan Strassle Rojas, **Imran Shah**, Jimena Martin Tempestti, Alessandro Veneziani, Brooks Lindsey. Intravascular Ultrasound Wall Shear Stress Imaging in Stented Coronary Arteries with Ultrafast Doppler. *Recently Accepted, Scientific Reports*. 2026
- David Molony, Lovia Hung, Michel Corban, Bill Gogas, Adrien Lefieux, **Imran Shah**, et. al. A randomized prospective study investigating the relationship between post-PCI wall shear stress and 12-month neointimal healing: The SHEAR-STENT study. *Cardiovascular Revascularization Medicine*, 2025. https://doi.org/10.1016/j.carrev.2025.11.001
- **Imran Shah,** David Molony, Adrien Lefieux, Kaylyn Crawford, Marina Piccinelli, Hanyao Sun, Don Giddens, Habib Samady, Alessandro Veneziani. Impact of the Stent Footprint on Endothelial Wall Shear Stress in Patient-Specific

Coronary Arteries: A Computational Analysis from the SHEAR-STENT Trial. *Computer Methods and Programs in Biomedicine*, 266:108762, 2025. https://doi.org/10.1016/j.cmpb.2025.108762

- **Imran Shah**, Milad Samaee, Atefeh Razavi, Fateme Esmailie, Lakshmi P. Dasi, Alessandro Veneziani. Reduced Order Modeling for Real-Time Stent Deformation Simulations of Transcatheter Aortic Valve Prostheses. *Annals of Biomedical Engineering*, 52(2):208-225, 2024. https://doi.org/10.1007/s10439-023-03360-5
- Sonali Kumar, David Molony, Sameer Khawaja, Kaylyn Crawford, Elizabeth Thompson, Olivia Hung, **Imran Shah**, et. al. Stent underexpansion is associated with high wall shear stress: a biomechanical analysis of the shear stent study. *The International Journal of Cardiovascular Imaging*, 39:1375-82, 2023. https://doi.org/10.1007/s10554-023-02838-6
- Fateme Esmailie, Atefeh Razavi, Breandan Yeats, Sri Krishna Sivakumar, Huang Chen, Milad Samaee, **Imran Shah**, Alessandro Veneziani, Pradeep Yadav, Vinod Thourani, Lakshmi P. Dasi. Biomechanics of Transcatheter Aortic Valve Replacement Complications and Computational Predictive Modeling. *Structural Heart*, 6(2), 2022. https://doi.org/10.1016/j.shj.2022.100032
- David Molony, Adrien Lefieux, Kaylyn Crawford, **Imran Shah**, et. al. A Comparison of Post-Stenting Wall Shear Stress Between Xience and Resolute: Interim Analysis from the Shear-Stent Trial. *Journal of the American College of Cardiology*, 79(9):828, 2022. https://doi.org/10.1016/S0735-1097(22)01819-8

Manuscripts in Preparation

- **Imran Shah,** Francesco Ballarin, Zhenglun A. Wei, Ajit P. Yoganathan, Lakshmi P. Dasi, Alessandro Veneziani. Real-Time Hemodynamic and Shape Optimization Simulations for Patient-Specific Fontan Surgical Planning via Reduced Order Modeling. *In Preparation.*
- **Imran Shah,** Francesco Ballarin, Zhenglun A. Wei, Habib Samady, Lakshmi P. Dasi, Alessandro Veneziani. Model Order Reduction for Shape Optimization in Cardiovascular Fluid Mechanics: Applications in Fontan Surgical Planning and Coronary Stent Design. *In Preparation.*
- **Imran Shah,** Francesco Ballarin, Alessandro Veneziani, Lakshmi P. Dasi. Real-Time Predictive Modeling of Patient-Specific TAVR Deployment Mechanics. *In Preparation.*

Selected Conference Proceedings

Published Abstracts:

- **Imran Shah,** Francesco Ballarin, Alessandro Veneziani, Vinod Thourani, Laksmi P. Dasi. TCT-1175 A Novel In-Silico Framework for Real-Time Predictive Modeling of Transcatheter Aortic Valve Replacement Mechanics. *Journal of the American College of Cardiology*, 86(17), 2025. https://doi.org/10.1016/j.jacc.2025.09.1393
- David Molony, Olivia Hung, Michel Corban, Bill D Gogas, **Imran Shah**, et. al. A Prospective OCT Study Investigating the Relationship Between Post-PCI Wall Shear Stress and 12-month Neointimal Healing: the SHEAR-STENT Study. *JACC*, 85(12). https://doi.org/10.1016/S0735-1097(25)02591-4
- **Imran Shah**, Sri Krishna Sivakumar, Francesco Ballarin, Vinod Thourani, Alessandro Veneziani, Lakshmi P. Dasi. TCT-206 Real-Time Predictive Modeling for Deployment of Transcatheter Aortic Valve Prostheses Using a Novel Reduced Order Model-Driven Computational Framework. *Journal of the American College of Cardiology*, 84(18), 2024. https://doi.org/10.1016/j.jacc.2024.09.242
- Courtney Ream, **Imran Shah**, Aniket Venkatesh, Brandon Walsh, Shantanu Bailoor, Lakshmi P. Dasi. Biomechanical Analysis of the J-Valve Bioprosthesis: A Computed tomography-Based Computational Study. *Journal of the American College of Cardiology*, 84(18), 2024. https://doi.org/10.1016/j.jacc.2024.09.1035
- **Imran Shah**, David Molony, Adrien Lefieux, Kaylyn Crawford, Hanyao Sun, Marina Piccinelli, Alessandro Veneziani, Habib Samady. TCT-606 Alterations in the Endothelial Wall Shear Stress Due to the Prescence of the Stent Footprint: Comparison Between Xience and Resolute Stent Platforms. *Journal of the American College of Cardiology*, 82(17), 2023. https://doi.org/10.1016/j.jacc.2023.09.617
- David Molony, Adrien Lefieux, Kaylyn Crawford, **Imran Shah**, et. al. A Comparison of Post-Stenting Wall Shear Stress Between Xience and Resolute: Interim Analysis from the Shear-Stent Trial. *Journal of the American College of Cardiology*, 2022. https://doi.org/10.1016/S0735-1097(22)01819-8
- David Molony, Juhwan Lee, Marina Piccinelli, … **Imran Shah**, et. al. TCT-CONNECT-283 Investigation of the Association of Wall Shear Stress and Fibrous Cap Thickness Assessed by Deep Learning. *Journal of the American College of Cardiology*, 2020. https://doi.org/10.1016/j.jacc.2020.09.300

Presentations:

- **Imran Shah**, Francesco Ballarin, Alessandro Veneziani, Vinod Thourani, Lakshmi P. Dasi. A Novel Reduced Order Model-Driven Paradigm for Real-Time Predictive Modeling of TAVR and Pre-Procedural Risk Assessment of Coronary Obstruction. *10th World Congress of Biomechanics*. July 11-15, 2026. Vancouver, Canada. Podium.

- **Imran Shah**, Francesco Ballarin, Alessandro Veneziani, Vinod Thourani, Lakshmi P. Dasi. Real-Tiem Predictive Modeling of Coronary Obstruction Following TAVR via Model Order Reduction. *Heart Valve Society Annual Meeting*. April 15-18, 2026. Chicago, IL, USA. Poster.
- **Imran Shah**, Francesco Ballarin, Alessandro Veneziani, Vinod Thourani, Lakshmi P. Dasi. A Novel *In-Silico* Framework for Real-Time Predictive Modeling of TAVR Mechanics. *Transcatheter Cardiovascular Therapeutics (TCT)*. October 25-28, 2025. San Francisco, CA, USA. Podium.
- **Imran Shah**, Francesco Ballarin, Zhenglun A. Wei, Alessandro Veneziani, Lakshmi P. Dasi. Accelerated Shape Optimization of Patient-Specific Fontan Surgical Planning Procedures. *10th International Biofluid Mechanics and Mechanobiology Symposium (IBMS 10)*. September 5-7, 2025. Irvine, CA, USA. Podium.
- **Imran Shah**, Francesco Ballarin, Zhenglun A. Wei, Lakshmi P. Dasi, Alessandro Veneziani. Real-Time Shape Optimization of Patient-Specific Fontan Surgical Planning Procedures. *ASME SB3C Summer Bioengineering Conference,* June 22-25, 2025. Santa Ana Pueblo, NM, USA. Podium.
- **Imran Shah**, Francesco Ballarin, Zhenglun A. Wei, Lakshmi P. Dasi, Alessandro Veneziani. Real-Time Shape Optimization of Patient-Specific Fontan Surgery via a Reduced Order Modeling Paradigm. *Georgia Scientific Computing Symposium*. February 2025. Atlanta, GA, USA. Poster.
- **Imran Shah**, Francesco Ballarin, Vinod Thourani, Alessandro Veneziani, Lakshmi P. Dasi. A Novel In-Silico Paradigm for Patient-Specific Predictive Modeling of Transcatheter Aortic Valve Deployment Mechanics. *Georgia Chapter of the American College of Cardiology*. November 22-24, 2024. Podium.
- **Imran Shah**, Sri Krishna Sivakumar, Francesco Ballarin, Vinod Thourani, Alessandro Veneziani, Lakshmi P. Dasi. Real-Time Predictive Modeling for Deployment of Transcatheter Aortic Valve Prostheses Using a Novel Reduced Order Model-Driven Computational Framework. *Transcatheter Cardiovascular Therapeutics (TCT).* October 27-30, 2024. Washington, D.C., USA. Podium.
- **Imran Shah**, Francesco Ballarin, Lakshmi P. Dasi, Alessandro Veneziani. Towards Real-Time Shape Optimization of the Total Cavopulmonary Connection using a Reduced Order Modeling Paradigm. *Single Ventricle Investigator Meeting*. October 17-19, 2024. Denver, CO, USA. Poster.
- **Imran Shah**, Francesco Ballarin, Lakshmi P. Dasi, Alessandro Veneziani. A Reduced Order Modeling Framework for Real-Time Shape Optimization of the Total Cavopulmonary Connection. *Reduced Order Modeling and Machine Learning for Large Eddy Simulations (ROM+ML4LES+)*. October 13-15, 2024. Atlanta, GA, USA. Podium.
- **Imran Shah**, David Molony, Adrien Lefieux, Kaylyn Crawford, Marina Piccinelli, Habib Samady, Alessandro Veneziani. Investigating Endothelial Wall Shear Stress Perturbations Due to the Coronary Stent Footprint. *8th International Conference on Computational and Mathematical Biomedical Engineering (CMBE)*. June 24-26, 2024. Arlington, VA, USA. Podium.
- **Imran Shah**, Sri Krishna Sivakumar, Francesco Ballarin, Vinod Thourani, Alessandro Veneziani, Lakshmi P. Dasi. Accelerated Stent Deployment Simulations via model Order Reduction for Predictive Modeling of Transcatheter Aortic Valve Replacement. *Summer Biomechanics, Bioengineering, and Biotransport Conference (SB3C)*. June 11-14, 2024. Lake Geneva, WI, USA. Podium.
- **Imran Shah**, Francesco Ballarin, Lakshmi P. Dasi, Alessandro Veneziani. Efficient Shape Optimization of the Total Cavopulmonary Connection via Hyper-Reduced Order Models and Free Form Deformation. *Summer Biomechanics, Bioengineering, and Biotransport Conference (SB3C)*. June 11-14, 2024. Lake Geneva, WI, USA. Podium.
- **Imran Shah**, Sri Krishna Sivakumar, Francesco Ballarin, Venkateshwar Polsani, Vinod Thourani, Alessandro Veneziani, Lakshmi P. Dasi. Real-Time Computational Modeling for Deployment of Transcatheter Aortic Valve Prostheses via Reduced Order Models. *Wallace H. Coulter Department of Biomedical Engineering BIO-STAR AI Symposium*. April 10-11, 2024. Atlanta, GA, USA. Poster.
- **Imran Shah**, Sri Krishna Sivakumar, Francesco Ballarin, Venkateshwar Polsani, Vinod Thourani, Alessandro Veneziani, Lakshmi P. Dasi. Real-Time Computational Modeling for Deployment of Transcatheter Aortic Valve Prostheses via Reduced Order Models. *American College of Cardiology (ACC) Annual Scientific Session*. April 6-8, 2024. Atlanta, GA, USA. Poster.
- **Imran Shah**, Francesco Ballarin, Lakshmi P. Dasi, Alessandro Veneziani. Accelerated Optimization of the Total Cavopulmonary Connection for Fontan Surgical Planning using Hyper-Reduced Order Models. *Georgia Scientific Computing Symposium*. February 2024. Atlanta, GA, USA. Poster.
- **Imran Shah**, David Molony, Adrien Lefieux, Kaylyn Crawford, Hanyao Sun, Marina Piccinelli, Alessandro Veneziani, Habib Samady. Alterations in the Endothelieal Wall Shear Stress Due to the Presence of the Stent Footprint: Comparison Between Xience and Resolute Stent Platforms. *Transcatheter Cardiovascular Therapeutics (TCT)*. November 2023. San Francisco, CA, USA. Podium.
- **Imran Shah**, Francesco Ballarin, Traian Iliescu, Omer San, Lakshmi P. Dasi, Alan Wei, Alessandro Veneziani. Real-Time Optimization of the Total Cavopulmonary Connection via Reduced Order Modeling. *Summer Biomechanics, Bioengineering, and Biotransport (SB3C) Conference*. June 4-8, 2023. Vail, Colorado, USA. Podium.

- **Imran Shah**, Francesco Ballarin, Alessandro Veneziani, Lakshmi P. Dasi. Towards Real-Time Predictive Modeling of Transcatheter Aortic Valve Replacement Procedures via Reduced Order Modeling. *Summer Biomechanics, Bioengineering, and Biotransport (SB3C) Conference.* June 4-8, 2023. Vail, Colorado, USA. Podium.
- **Imran Shah**, David Molony, Kaylyn Crawford, Adrien Lefieux, Alessandro Veneziani, Habib Samady. Impact of the Coronary Stent Footprint on Wall Shear Stress in Patient-Specific Arteries – Analysis from the SHEAR-STENT Trial. *Summer Biomechanics, Bioengineering, and Biotransport Conference*. June 4-8, 2023. Vail, Colorado, USA. Poster.
- **Imran Shah**, Milad Samaee, Alessandro Veneziani, Lakshmi P. Dasi. Towards Real-Time Predictive Modeling of Transcatheter Aortic Valve Replacement Procedures via Reduced Order Modeling. *Heart Valve Society Annual Scientific Meeting*. March 2023.
- **Imran Shah**, Huang Chen, Lakshmi P. Dasi, Ajit P. Yoganathan, Zhenglun A. Wei, Traian Iliescu, Omer San, Alessandro Veneziani. Towards Real-Time Shape Optimization of the Total Cavopulmonary Connection in Fontan Surgical Planning. *SIAM Conference on Computational Science and Engineering (CSE)*. Feb 26 – Mar 3, 2023. Amsterdam, The Netherlands. Podium.
- **Imran Shah**, David Molony, Adrien Lefieux, Kaylyn Crawford, Alessandro Veneziani, Habib Samady. Impact of the Stent Footprint on Wall Shear Stress Distribution in Patient-Specific Coronary Arteries – Analysis from the SHEAR-STENT Trial. *Georgia Chapter of the American College of Cardiology*. November 17-19, 2022. Greensboro, GA, USA. Podium.
- **Imran Shah**, Milad Samaee, Atefeh Razavi, Fateme Esmailie, Alessandro Veneziani, Lakshmi P. Dasi. Reduced Order Modeling Framework for Rapid Simulations of Transcatheter Aortic Valve Replacement Procedures. *European Solid Mechanics Conference (ESMC)*. 2022. Podium.
- **Imran Shah**, Huang Chen, Zhenglun A. Wei, Lakshmi P. Dasi, Ajit P. Yoganathan, Traian Iliescu, Omer San, Alessandro Veneziani. Reduced Order Modeling for Real-Time Shape Optimization of the Total Cavopulmonary Connection in Fontan Surgical Planning. *World Congress of Biomechanics*. 2022. Podium
- **Imran Shah**, Milad Samaee, Atefeh Razavi, Fateme Esmailie, Alessandro Veneziani, Lakshmi P. Dasi. Reduced Order Modeling Framework for Rapid Simulations of Transcatheter Aortic Valve Replacement Procedures. *Summer Biomechanics, Bioengineering, and Biotransport Conference*. 2022. Poster.
- **Imran Shah**, Huang Chen, Zhenglun A. Wei, Lakshmi P. Dasi, Ajit P. Yoganathan, Traian Iliescu, Omer San, Alessandro Veneziani. Reduced Order Modeling for Real-Time Shape Optimization of the Total Cavopulmonary Connection in Fontan Surgical Planning. *Summer Biomechanics, Bioengineering, and Biotransport Conference*. 2022. Poster.
- **Imran Shah**, Milad Samaee, Atefeh Razavi, Alessandro Veneziani, Lakshmi P. Dasi. Developing a Reduced Order Modelling Methodology for Rapidly Simulating Transcatheter Aortic Valve Replacement Procedures. *Summer Biomechanics, Bioengineering, and Biotransport Conference.* 18 June 2021. Poster.

Patents

- Lakshmi Prasad Dasi, Breandan Andre Butler Yeats, Huang Chen, Fateme Esmailie, Atefeh Razavi, Imran Shah, Sri Krishna Sivakumar, Alessandro Veneziani. Systems and methods for modeling risk of transcatheter valve deployment. U.S. Patent Application No. 18/560,210.

Teaching Experience

Georgia Institute of Technology, Department of Biomedical Engineering

- **Graduate Student Research Mentor,** Cardiovascular Fluid Mechanics Lab – Fall 2023 – Current
- **Undergraduate Student Research Mentor**, Cardiovascular Fluid Mechanics Lab – Summer 2025 – Current
- **Graduate Teaching Assistant,** BMED 3410: Introduction to Biomechanics – Fall 2021, Spring 2022
- **Teaching Assistant**, BMED 2310: Introduction to Biomedical Engineering Design – Summer 2017, Fall 2017
- **Teaching Assistant**, BMED 3520: Biomedical Systems and Modeling – Fall 2018
- **Undergraduate Mentor**, Biomedical Engineering Student Advisory Board (BMED SAB)

Georgia State University

- **Tutor**, CHEM 2400: Organic Chemistry – 2016
- **Tutor**, MATH 3260: Differential Equations – 2016

Skills

- **Programming & Design:** Python, MATLAB, C++, FreeFem++, SolidWorks, Inventor, Rhino 3D, Geomagic Studio
- **CFD/FEA:** FEniCS, RBniCS, Ansys Workbench Platform, Fluent, ABAQUS, ICEM VMTK, Gmsh, Netgen, ParaView, Finite Element Analysis, Finite Differences
- **Fabrication:** 3D Printing, Laser Cutting, Mechanical Testing (Tensile), Rapid Prototyping, Soldering, Band Saw
- **Lab Techniques:** Mammalian Cell Culture, Apoptosis Assays, Cytotoxicity Assays, Phase Contrast Microscopy

Honors & Awards

- Georgia Chapter of the American College of Cardiology Governor's Award for Excellence in Research 2024
- CREATE-X Startup Launch Finalist – AQUApro: Data-driven Analytics of Stormwater Infrastructure 2024
- American Heart Association Predoctoral Fellowship 2024 – 2025
- Emory Global Health Institute/Georgia Institute of Technology Global Health Hackathon – 1st Place 2023
- Emory University GHA 2040 Healthcare Futuring Competition – 1st Place 2023
- Georgia Chapter of the American College of Cardiology Governor's Award for Excellence in Research 2022, 2024

Professional Memberships

- Society for Industrial and Applied Mathematics 2022 – 2025
- American Heart Association 2021 – 2025
- Biomedical Engineering Society 2016 – 2019

Relevant Coursework

- Turbulent Flows – Magnetic Resonance Imaging – Machine Learning – Numerical Partial Differential Equations – Biofluid Mechanics – Computational Fluid Mechanics – Biotransport – Drug Design, Development, and Delivery – Cancer Biology and Biotechnology – RNA Biology and Biotechnology